

Secret-Subspace Recovery in MAYO via Linearization of Errors from a Single Fault

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Abstract

We present a fault injection attack against MAYO that, from a single faulty execution, enables the recovery of structural information about the secret. We consider a simple fault model: a controlled perturbation in a single *oil* coordinate of a signature block, which induces an error $e \in \mathcal{O}$ (secret subspace) with known *oil* part. We show that the observable mismatch in verification, $\Delta t = P^*(s') - t$, can be expressed exactly as the image of e under a publicly derivable linear operator $\mathcal{L}$, obtained by expanding P^* and using (i) the bilinearity of the differential P' in characteristic 2 and (ii) the key property $P(u) = 0$ for all $u \in \mathcal{O}$. This linearization makes it possible to separate *vinegar/oil* coordinates and to reduce the recovery of the unknown component e_V to the solution of a linear system over $\mathbb{F}_q$, under generic full-rank conditions for typical parameters. Once e is recovered, the faulty signature can be corrected and, more importantly, a nonzero vector of the secret subspace is obtained, which serves as a starting point to scale to key recovery via known *oil space* reconstruction techniques. We further discuss the practical feasibility when the exact position/value of the fault is unknown, showing that a bounded search over $k \cdot o$ positions and $q - 1$ values keeps the cost low for the official parameter sets, and that the attack is also applicable to the randomized variant of MAYO.

1 Introduction

The National Institute of Standards and Technology (NIST) has recently concluded its post-quantum cryptography (PQC) standardization process, selecting schemes designed to withstand threats arising from quantum computing. Among the adopted standards are the digital signature schemes ML-DSA [Nat24a] and SLH-DSA [Nat24b]. With the aim of broadening the range of viable alternatives in PQC, NIST launched in September 2022 a second call focused on identifying new signature algorithms that offer different security assumptions and performance trade-offs from those of ML-DSA and SLH-DSA.

In this context, the MAYO signature scheme [Beu22, BCC+25] has emerged as one of the most relevant candidates selected by NIST for consideration in the new PQC signature call. MAYO belongs to the family of multivariate schemes, whose security relies on the computational hardness of solving systems of nonlinear polynomial equations over finite fields. However, beyond theoretical guarantees, it is essential to assess its practical robustness against adversaries with physical capabilities. The literature has repeatedly shown that side-channel attacks and, in particular, fault injection attacks can compromise implementations of post-quantum schemes, highlighting a persistent gap between theoretical resistance and security in real-world deployments. Therefore, rigorous analysis of susceptibility to physical attacks and the design of effective countermeasures should be regarded as fundamental components of any PQC adoption strategy.

In recent years, fault injection attacks against multivariate schemes have received increasing attention in the literature, with a particular focus on the MAYO scheme [AMSU24, ABC+25,

Zho25, BDK25]. These works consider different attack models and fault scenarios, and analyze how perturbations during the signing process can lead to the exposure of secret information.

In this article, we study a fault injection scenario in MAYO in which perturbations on internal variables during the signing process give rise to additional algebraic relations. From these relations, we show that it is possible to recover information about the secret key, even when the randomized variant of the algorithm is employed.

Contributions. To the best of our knowledge, this is the first work that presents a *single-fault* attack against *randomized* MAYO that, without requiring any correct reference signature, enables the recovery of the secret oil subspace, where the injected fault is not an instruction-skip or loop-abort fault.

- We propose a simple and realistic *fault model* in which a controlled perturbation affects a single *oil* coordinate of one of the signature blocks, before that block is combined with the secret subspace.
- We derive an *explicit linearization* of the observable mismatch during verification as a function of the induced error, exploiting the bilinear structure of the scheme.
- We show that the *full error* can be recovered by solving a linear system over $\mathbb{F}_q$, leveraging the fact that the *oil* component of the error is known under the fault model.
- We analyze how *error recovery* leads to the extraction of a nonzero vector of the secret subspace and discuss how, from this vector, one can scale to *key recovery* using existing results in the literature on UOV schemes, in particular the procedure of Pébereau [Peb24].

2 Preliminaries and Scheme Description

Let $\mathbb{F}_q$ be a finite field of characteristic 2 (in the proposed parameter sets one uses $q = 16$). The scheme is parameterized by (n, m, o, k, q) . The public key defines a homogeneous quadratic map

$$P : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q^m, \quad P(x) = (p_0(x), \dots, p_{m-1}(x)),$$

with the trapdoor property that there exists a secret linear subspace $\mathcal{O} \subset \mathbb{F}_q^n$ of dimension o on which P vanishes: $P(u) = 0$ for all $u \in \mathcal{O}$.

In Oil–Vinegar-type implementations it is natural to partition a vector $x \in \mathbb{F}_q^n$ into two blocks

$$x = (w \| x_O), \quad w \in \mathbb{F}_q^{n-o}, \quad x_O \in \mathbb{F}_q^o,$$

and to represent the secret subspace in systematic form by means of a matrix $O \in \mathbb{F}_q^{(n-o) \times o}$:

$$\mathcal{O} = \{(O\alpha \| \alpha) : \alpha \in \mathbb{F}_q^o\}. \quad (1)$$

The secret data is essentially O (or an equivalent encoding); the public key determines P .

The MAYO design introduces a *public amplification* that transforms P into a new map

$$P^* : \mathbb{F}_q^{kn} \rightarrow \mathbb{F}_q^m, \quad P^*(x_0, \dots, x_{k-1}) = \sum_{i=0}^{k-1} E_{ii} P(x_i) + \sum_{0 \leq i < j \leq k-1} E_{ij} P'(x_i, x_j),$$

where $\{E_{ij}\} \subset \mathbb{F}_q^{m \times m}$ are fixed public matrices and

$$P'(x, y) := P(x + y) - P(x) - P(y) \quad (2)$$

is the differential of P . In characteristic 2, P' is a symmetric bilinear map.

2.1 Key Generation Algorithm

Key generation produces a public instance P with Oil–Vinegar structure and a secret subspace $\mathcal{O}$ such that $P(u) = 0$ for all $u \in \mathcal{O}$. The secret key encodes a basis of $\mathcal{O}$ (equivalently, the matrix O of (1) in systematic form), while the public key encodes the coefficients of P and the matrices E_{ij} (or seeds from which they are expanded).

2.2 Signing Algorithm

Given a message M , the signer derives a target $t \in \mathbb{F}_q^m$ via a hash with **salt**. It then randomly samples k base vectors $v_0, \dots, v_{k-1} \in \mathbb{F}_q^n$ and searches for corrections $o_0, \dots, o_{k-1} \in \mathcal{O}$ such that

$$P^*(v_0 + o_0, \dots, v_{k-1} + o_{k-1}) = t. \quad (3)$$

The key point is that (3) reduces to a linear system in the ko free coordinates that parameterize $(o_0, \dots, o_{k-1})$. To see this, one expands using (2):

$$P(v_i + o_i) = P(v_i) + P'(v_i, o_i) + P(o_i) = P(v_i) + P'(v_i, o_i),$$

since $P(o_i) = 0$ by definition of $\mathcal{O}$. Similarly, when $P'(v_i + o_i, v_j + o_j)$ appears in the cross part of P^* , all terms involving o_i, o_j enter through P' , which is bilinear, and moreover $P'(o_i, o_j) = P(o_i + o_j) - P(o_i) - P(o_j) = 0$ because $o_i, o_j, o_i + o_j \in \mathcal{O}$. Therefore, (3) becomes a linear system in the unknowns describing $o_0, \dots, o_{k-1}$. A solution yields a signature

$$s = (s_0, \dots, s_{k-1}), \quad s_i = v_i + o_i \in \mathbb{F}_q^n,$$

together with the **salt**.

2.3 Verification Algorithm

The verifier recomputes the target $t \in \mathbb{F}_q^m$ from (M, salt) and the public key, and accepts if

$$P^*(s_0, \dots, s_{k-1}) = t. \quad (4)$$

In efficient implementations, P^* is evaluated as a linear combination of bilinear products induced by P' and of quadratic evaluations of P weighted by the matrices E_{ij} .

3 Related Work

3.1 Physical security of MAYO: overview and relevant attacks

UOV-based signature schemes, including MAYO, have attracted increasing interest from the perspective of physical security, as their practical implementations introduce attack surfaces not captured by the classical algorithmic model. In particular, *side-channel analysis* (SCA) and *fault injection* (FI) attacks can perturb the signing process and lead to the exposure of internal information. An overview of these vectors and of the official variants of MAYO can be found in [ACK25, BCC+25].

We next summarize the FI attacks most relevant to this work, highlighting the fault injection point and the conditions under which they enable the extraction of secret information.

FI on vinegar sampling. Aulbach *et al.* [AMSU24] present the first *end-to-end* FI attack on MAYO. The fault is induced during the sampling of the vinegar vectors, via an *instruction skip* or *loop abort*, so that one or more vinegars are not generated. Under a known-initialization (*zeroing*) scenario, the faulty signature directly reveals a vector of the oil subspace; alternatively, in a differential scenario, the comparison of related signatures allows the vinegar part to be canceled and information about the *oil space* to be isolated. The attack may require a single faulty execution and has been experimentally validated using *clock glitching*.

FI on the derivation of the *vinegar seed*. Jendral and Dubrova [JD25] propose several *single-execution* FI attacks against MAYO implementations, targeting the derivation of the seed that generates the vinegars. By means of *instruction skips* in the implementation of SHAKE256/Keccak or in buffer initialization, the attacker forces the seed to be constant or predictable from public information, which enables secret key recovery from a single faulty signature. These attacks apply to both the deterministic and the randomized variants of MAYO.

FI to obtain related valid signatures. Zhong [Zho25] introduces a theoretical FI framework that exploits the existence of free variables in the linear system solved during MAYO signing. By inducing faults that modify the pseudorandom vector r , the attacker can obtain multiple valid signatures for the same message and *salt*. Comparing these signatures allows information about the oil subspace to be extracted. The work highlights that randomization hinders this scenario by reducing the probability of related signatures.

Practical FI on randomized MAYO with a verification oracle. Abdelmonem *et al.* [ABC+25] present a practical FI attack specifically targeting randomized MAYO. The fault is injected during the operation that combines vinegar and oil in the signature, causing partial omission of the oil term. The attack is completed using *signature correction* techniques, leveraging the verifier as an oracle to progressively reconstruct the *oil space*.

FI on deterministic variants via re-signing. Bauer, De Santis, and Koleci [BDK25] study FI attacks on deterministic UOV schemes, including deterministic MAYO. By obtaining two signatures of the same message (one correct and one faulty), the reuse of the same vinegar vector allows this component to be canceled and a vector of the oil subspace to be revealed. The attack explicitly exploits the determinism of the scheme and does not directly apply to randomized variants.

4 Threat Model and Fault Model

4.1 Adversary Capabilities

We assume an adversary with access to the signing device who can:

- request signatures on messages that are known to the adversary, with adaptive access to a legitimate signing oracle;
- induce, in a controlled manner, a transient and localized fault during a single execution of the signing algorithm;
- observe the signature produced by the faulty execution and know the signed message and the associated public key, so that they can locally run the public verification algorithm and compute the observable mismatch introduced by the fault.

4.2 Fault Model

We assume a fault injection attack that alters an *oil* coordinate of the block x_O in the first signed component. Without loss of generality, we assume that the fault affects block 0 and coordinate 0 of the oil vector, so that the injected oil correction is

$$\delta e_0 \in \mathbb{F}_q^o, \quad \delta \in \mathbb{F}_q^\times, \quad e_0 = (1, 0, \dots, 0). \quad (5)$$

In the systematic representation (1), this induces an error vector in the full space

$$e := (O(\delta e_0) \parallel \delta e_0) \in \mathbb{F}_q^n, \quad (6)$$

which belongs to the secret subspace $\mathcal{O}$. The faulty signature then has the form

$$s' = (s'_0, s_1, \dots, s_{k-1}), \quad s'_0 = s_0 + e, \quad (7)$$

while all other blocks remain unchanged.

4.3 Fault Observation

The immediate consequence is that the verification equation (4) generally no longer holds:

$$P^\star(s') \neq t.$$

Since the attacker has access to (M, salt) and to the faulty signature s' , they can also compute the *observable mismatch*

$$\Delta t := P^\star(s') - t \in \mathbb{F}_q^m. \quad (8)$$

If the original signature s was valid, then $t = P^\star(s)$ and

$$\Delta t = P^\star(s') - P^\star(s), \quad (9)$$

that is, Δt coincides exactly with the change induced by the error e .

5 Attack Description

The attack consists of two phases:

1. **Recovery of the induced error.** From Δt , a linear system is constructed to recover e (in particular its *vinegar* part), exploiting the fact that the oil part of the error is known under the fault model.
2. **Post-processing for secret key recovery.** Since $e \in \mathcal{O}$, a nonzero vector of the secret subspace serves as a witness to attempt reconstruction of an equivalent representation of the secret key (for example, a matrix O).

5.1 Error Recovery: Linearization via the Public Map $P^\star$

The goal is to reconstruct e (or, more precisely, its vinegar part) from Δt , leveraging the fact that the attacker knows *which* oil coordinate was perturbed, that is, they know (5).

5.1.1 Step 1: Expanding $P^\star(s') - P^\star(s)$ and isolating the changing terms

By (7), the only block that changes is s_0 . Substituting into the definition of $P^\star$, all summands that do not involve s_0 cancel out in (9). We obtain

$$\Delta t = E_{00}(P(s_0 + e) - P(s_0)) + \sum_{j=1}^{k-1} E_{0j}(P'(s_0 + e, s_j) - P'(s_0, s_j)). \quad (10)$$

In the second term, the bilinearity of P' implies

$$P'(s_0 + e, s_j) - P'(s_0, s_j) = P'(e, s_j),$$

so the cross terms contribute as a sum of linear terms in e .

In the first term, we use the general identity for homogeneous quadratic maps:

$$P(x + e) - P(x) = P'(x, e) + P(e). \quad (11)$$

Here the decisive structural condition appears: since $e \in \mathcal{O}$, we have $P(e) = 0$. Therefore,

$$P(s_0 + e) - P(s_0) = P'(s_0, e).$$

Moreover, in characteristic 2 one has $P'(e, e) = 0$ for all e , and hence

$$P'(s_0, e) = P'(s_0 + e, e).$$

This allows the diagonal term to be rewritten *as a function of $s'_0 = s_0 + e$* , which is known to the attacker. Substituting, we obtain a fully observable expression:

$$\Delta t = E_{00} P'(s'_0, e) + \sum_{j=1}^{k-1} E_{0j} P'(e, s_j). \quad (12)$$

Equation (12) is the starting point of the recovery.

5.1.2 Step 2: Why (12) is a linear relation in e

For each fixed $y \in \mathbb{F}_q^n$, the map

$$\Phi_y : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m, \quad \Phi_y(x) := P'(x, y),$$

is linear in x because P' is bilinear. The same holds for $x \mapsto P'(y, x)$. Moreover, left multiplication by $E_{ij} \in \mathbb{F}_q^{m \times m}$ is linear on $\mathbb{F}_q^m$. Therefore, the right-hand side of (12) is a linear combination of linear maps applied to e . Consequently, there exists a unique linear transformation

$$\mathcal{L} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m \quad \text{such that} \quad \Delta t = \mathcal{L}(e), \quad (13)$$

with

$$\mathcal{L}(e) := E_{00} P'(s'_0, e) + \sum_{j=1}^{k-1} E_{0j} P'(e, s_j). \quad (14)$$

Observe that, since P' is bilinear and symmetric, the terms defining $\mathcal{L}$ can equivalently be rewritten by always placing the error e as the first argument. In particular, symmetry implies that $P'(s'_0, e) = P'(e, s'_0)$, and bilinearity guarantees that, for each fixed vector y , the map $e \mapsto P'(e, y)$ is linear. In this way, all summands composing $\mathcal{L}(e)$ can be uniformly interpreted as evaluations of linear maps in the variable e , which sets the stage for the explicit analysis of the linear dependence on the coordinates of e in the next step.

5.1.3 Step 3: Separating vinegar/oil coordinates and exploiting the fact that the oil part of the error is known

We decompose the error as

$$e = (e_V \| e_O), \quad e_V \in \mathbb{F}_q^{n-o}, \quad e_O \in \mathbb{F}_q^o. \quad (15)$$

Under the fault model (5), the attacker knows

$$e_O = \delta e_0. \quad (16)$$

The vinegar part e_V is unknown (and, in the systematic model, coincides with $O(\delta e_0)$).

Since $\mathcal{L}$ is linear, it also decomposes uniquely into two linear contributions, one associated with the first $n - o$ coordinates and the other with the last o coordinates:

$$\mathcal{L}(e_V \| e_O) = \mathcal{L}_V(e_V) + \mathcal{L}_O(e_O), \quad (17)$$

where $\mathcal{L}_V : \mathbb{F}_q^{n-o} \rightarrow \mathbb{F}_q^m$ and $\mathcal{L}_O : \mathbb{F}_q^o \rightarrow \mathbb{F}_q^m$ are linear. Substituting (17) into (13) and using (16) yields:

$$\mathcal{L}_V(e_V) = \Delta t - \mathcal{L}_O(\delta e_0). \quad (18)$$

Equation (18) is a linear system that determines e_V from observable quantities: Δt is computed via verification, and $\mathcal{L}_O(\delta e_0)$ can be computed because both δe_0 and $\mathcal{L}$ are publicly evaluable (they depend on P^* , on s' , and on the matrices E_{ij}).

In the main MAYO parameter sets, it typically holds that $m = n - o$. In that case, (18) becomes a square system in $n - o$ unknowns, and under the generic full-rank condition it has a unique solution. If $m \neq n - o$, the system can still be solved by Gaussian elimination over $\mathbb{F}_q$; recovery succeeds whenever the system is consistent and the rank of $\mathcal{L}_V$ is sufficient to fix a solution.

In what follows, we *restrict our attention to the case in which the linear system (18) has full rank*, so that the solution for e_V is unique. Note that this condition depends exclusively on the signature block affected by the fault and on the public data involved in the construction of the linear map $\mathcal{L}_V$, and not on the specific *oil* coordinate nor on the value of the error δ . In particular, there may exist valid instances of the fault model for which, even when the hypothesis on the position and the value of the error is correct, the linear map $\mathcal{L}_V$ turns out to be singular and the system does not determine e_V uniquely. In such cases, the information contained in Δt is not sufficient to uniquely identify the induced error, and the described recovery procedure does not lead to an unambiguous solution. Since our goal is to recover the actual error in a deterministic and unambiguous manner, we discard these degenerate instances and consider only those for which $\mathcal{L}_V$ has full rank, which corresponds to the generic behaviour expected for the MAYO parameter sets.

Once e_V has been recovered, the full error is determined by (15) and (16). The faulty signature is then corrected by replacing s'_0 with $s'_0 - e$, leaving all other blocks unchanged. If the affected coordinate or the value of the error δe_0 are not known exactly, each hypothesis can be verified by checking whether the corrected signature satisfies the verification equation with the public data. If so, the correct error has been identified; otherwise, the process is repeated by solving again the linear system under a different hypothesis on the position or value of the fault.

6 Post-Processing: From Error Recovery to Key Recovery

The previous analysis shows that, from a single faulty signature, the error vector e can be recovered. Knowing e allows, if desired, the correction of the affected block and the production of a signature that satisfies $P^*(s) = t$. However, the central point is not to “fix” a particular signature, but rather to exploit the structural information revealed by the fault: the induced error is not an arbitrary vector of $\mathbb{F}_q^n$, but an element of the secret subspace $\mathcal{O}$.

Indeed, under the considered fault model (injection into an oil coordinate), the error takes the form

$$e = (O\alpha \parallel \alpha), \quad \alpha \in \mathbb{F}_q^o \setminus \{0\},$$

so that $e \in \mathcal{O}$ and, moreover, $e \neq 0$. This implies that, after the linearization phase via P^* and the solution of the linear system, the attacker obtains *a nonzero vector of the secret subspace*.

From this point, the natural objective becomes the full recovery of the secret matrix O , that is, of a systematic representation of the subspace $\mathcal{O}$. In particular, there exists a line of results showing that, in Oil–Vinegar-type schemes, having access to a single nonzero vector of the secret subspace, together with the public key, may be sufficient to reconstruct the entire oil structure and obtain an equivalent secret key.

A direct reference for this post-processing is the work *One Vector to Rule Them All: Key Recovery from One Vector in UOV Schemes* [Peb24], which describes how to transform a witness

$e \in \mathcal{O}$ and the public description of the quadratic map into an effective procedure for recovering the oil matrix.

7 Comments on the Practicality of the Attack

The form of the above equations is independent of the parameter set. What changes across variants are (i) the dimension o of the oil block, (ii) the number of blocks k , and (iii) the relation between m and $n - o$, which determines whether (18) is square or overdetermined. The assumption “fault in $x_0[0]$ ” entails no loss of generality: a fault in any oil coordinate and any block i^* is obtained by substituting the indices $0 \mapsto i^*$ and $e_0 \mapsto e_\ell$ (the corresponding canonical vector) in (5)–(18), and the linearization argument remains unchanged because it relies solely on the bilinearity of P' and on the vanishing of P over $\mathcal{O}$.

Moreover, even if the attacker does not know *which* error was induced nor *where* (block/position) it occurred, it suffices to consider an exhaustive search over both quantities. Under the model where the fault affects *a single* oil coordinate $x_{i^*}[\ell]$ (with $i^* \in \{0, \dots, k-1\}$ and $\ell \in \{0, \dots, o-1\}$), the number of candidate positions is $k \cdot o$. The induced error lives in $\mathbb{F}_q$ and, excluding the trivial case, there are $q - 1$ possible nonzero values (for MAYO, $q = 16$ and thus $q - 1 = 15$). Consequently, the brute-force cost to simultaneously identify the affected coordinate and the error value is

$$N_1 = (k \cdot o)(q - 1) = k \cdot o \cdot 15 \text{ trials.} \quad (19)$$

For the official parameter sets, this yields:

- **MAYO1** ($k = 10, o = 8$): $N_1 = 15 \cdot 10 \cdot 8 = 1200$.
- **MAYO2** ($k = 4, o = 17$): $N_1 = 15 \cdot 4 \cdot 17 = 1020$.
- **MAYO3** ($k = 11, o = 10$): $N_1 = 15 \cdot 11 \cdot 10 = 1650$.
- **MAYO5** ($k = 12, o = 12$): $N_1 = 15 \cdot 12 \cdot 12 = 2160$.

This would require solving the linear system in e and correcting the error in the faulty signature in order to check whether the correct error has been found, using the message, the public key, and the verification algorithm.

If the fault were to affect more than one oil coordinate (e.g., t coordinates), the search would grow exponentially: there are $\binom{ko}{t}$ ways to choose the positions and $(q - 1)^t$ possible error patterns, so the number of hypotheses scales as

$$N_t = \binom{ko}{t} (q - 1)^t, \quad (20)$$

which quickly becomes intractable as t increases.

Note on degenerate instances (non-full rank). The described recovery procedure assumes that, for the affected block i^* , the linear map $\mathcal{L}_V$ (equivalently, the matrix associated with the system in (18)) has full rank, so that e_V is uniquely determined. However, for a given concrete signature it may happen that, even under the *single-oil-coordinate fault* model, the instance associated with a particular block i^* yields a singular map $\mathcal{L}_V$ (that is, $\text{rank}(\mathcal{L}_V) < n - o$). In this case, even if the hypothesis on the coordinate ℓ and the error value δ is correct, the information contained in Δt does not allow the induced error to be deterministically identified, and recovery may fail for that specific execution. When the fault position is unknown, a brute-force attack must then iterate over all hypotheses (i^*, ℓ, δ) without being able to stop early; by contrast, if the attacker knew *a priori* the affected block, they could directly abort the search for that signature upon detecting that the corresponding system does not have full rank.

8 Countermeasures

The described attack exploits two facts: (i) a fault injection succeeds in introducing an error $e \in \mathcal{O}$ whose *oil* part is (partially) known or bounded, and (ii) the observable mismatch during verification linearizes as $\Delta t = \mathcal{L}(e)$, allowing recovery of e_V by solving a linear system and, subsequently, obtaining a nonzero witness of $\mathcal{O}$. Consequently, natural countermeasures fall into two categories: (a) preventing the adversary from obtaining a usable faulty signature, or (b) making the assumed fault model unstable (via detection and/or randomization).

8.1 Internal Verification

A direct defense is to internally check the verification equation before releasing the signature. Given a candidate s' , the signer can evaluate $P^*(s')$ and only output the signature if $P^*(s') = t$. This measure blocks the use of *invalid* signatures such as those yielding $\Delta t \neq 0$ in (8), since the attacker never gets to observe the pair $(s', \Delta t)$. This type of internal verification has also been considered in hardware implementations of MAYO as a countermeasure against fault injection attacks, where faulty executions are discarded before the signature is exposed to the outside [SMA+24].

8.2 Redundancy, Memory Coherence, and Protection of Sensitive Variables

The assumed fault model alters an *oil* coordinate in one of the signature blocks before the *oil* variables are combined with the secret subspace, which requires the faulty value to propagate undetected from the solution of the linear system to subsequent computations. To harden this point, redundancy and memory-coherence techniques can be employed, such as recomputing the *oil* variables obtained from the linear system and checking their consistency before use, or verifying the system residual to ensure that the solution is correct. Once validated, these variables can be protected via duplication and coherence checks during subsequent computations, forcing the adversary to introduce additional faults to simultaneously bypass the checks and thereby increasing the practical difficulty of the attack.

8.3 Specific Mitigation: Preventing the Fault from Inducing an Error with a “Knowable” Oil Part

In the attack derivation, the decisive step is treating e_O as known (or, failing that, enumerating it at low cost) in order to turn (13) into the system (18). Since the attacker can recover the fault position by brute force whenever the error affects a single oil coordinate, effective countermeasures do not consist in hiding that position, but rather in preventing the fault from inducing a structurally simple error.

In particular, the following measure can be considered:

- *Diffusion of the error across multiple coordinates*: designing the internal representation of signature blocks such that a local corruption propagates to multiple oil coordinates (or to a mixture of *vinegar* and *oil* coordinates). In that case, the hypothesis space grows combinatorially and exhaustive search over the errors quickly becomes infeasible.

This measure aims to break the assumption of (5) of a *single isolated oil error*, which is what enables efficient linearization and recovery of the secret subspace.

9 Conclusions

In this work, we present a fault injection attack against MAYO in which a controlled error induced during the signing process causes the resulting error in the signature to belong to the

secret *oil* subspace. The attack relies on a simple and realistic fault model, consisting of the perturbation of a single *oil* coordinate in one of the signed blocks. We show that the observable mismatch during verification can be expressed as the image of this error through an explicit and publicly evaluable linear operator, obtained by exploiting the bilinearity of the differential of the public map and the fact that vectors in the *oil* subspace vanish under the quadratic map, a basic structural property of UOV schemes. This linearization makes it possible to reduce the recovery of the induced error to the solution of a linear system over $\mathbb{F}_q$, enabling the extraction, from a single faulty signature, of a nonzero vector of the secret subspace, even when the randomized variant of MAYO is employed.

Beyond the correction of faulty signatures, the central result is that obtaining such a vector directly provides a sufficient witness to scale to full secret key recovery using techniques already known for UOV schemes. Consequently, this work highlights that fault injection attacks can compromise the security of MAYO at the private-key level without requiring strong assumptions about determinism, memory reuse, or weaknesses in auxiliary primitives, significantly broadening the scope of previously known physical attacks.

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